

**Homework 4**  
**35 points**

1. Consider  $V = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 1\}$ .

Is  $V$  a valid vector space? **10 points**

*Show valid steps and reasoning*

2. Consider

$B = \{(1,0,1), (0,1,1), (1,1,0)\}$  as a basis for  $\mathbb{R}^3$ .

Find the coordinates of the vector  $(2,3,1)$  (*currently in standard basis*) with respect to  $B$ . **5 points**

3. Let

$$v_1 = (1, 2, 1), v_2 = (2, 1, 3).$$

Is the vector  $w = (3, 3, 4)$  in  $\text{Span}\{v_1, v_2\}$ ? **5 points**

*Show valid steps*

4. Check if the set:

$v_1 = (1,1,0), v_2 = (0,1,1), v_3 = (1,2,1)$  is linearly independent. **5 points**

5. Let  $V = \mathbb{R}^2$

We define vector addition in  $V$  as follows:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2 + 1, y_1 + y_2).$$

Is  $V$  a valid vector space? **10 points**

***Show valid steps and reasoning***